Unit 2 Pre-Test

Functions & Linear Functions

Standards assessed: NC.M1.F-IF.1 · F-IF.2 · F-IF.4 · F-IF.5 · F-IF.6 · F-IF.7 · A-CED.2 · F-LE.5

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.M1.F-IF.1 · NC.M1.F-IF.2 · NC.M1.F-IF.4 · NC.M1.F-IF.6 · NC.M1.A-CED.2 Recognize functions, use f(x) notation, and read lines.

• *I can tell if a relation is a function, evaluate f(x), read the slope and y-intercept, and write y = mx + b.*

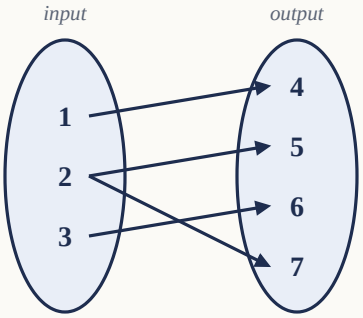
Answer each question. Use the picture to help you.

How to type your answers

number `-2` point `(0, 1)` equation `y=3x-2` interval `-3<=x<=4`

1. *Function?*

Is this relation a function? Write yes or no.



Answer: _____

`answer:` no (input 2 has two outputs)

2. *Evaluate*

f(x) = 3x − 2. Find f(4).

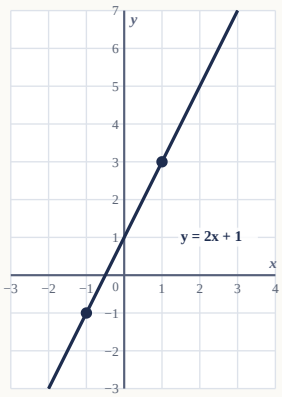


f(4) = _____

`answer:` 10

3. *Key feature*

Use the graph. What is the **y-intercept**?



Answer:

answer: 1 (the point (0, 1))

4. *Rate of change*

Use the table. What is the rate of change (slope)?

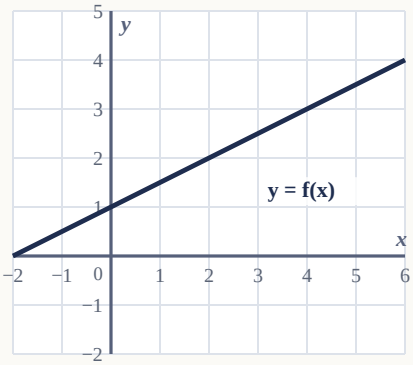
x	0	1	2	3
y	7	5	3	1

Answer:

answer: −2

5. *Read the graph*

The graph shows y = f(x). Find f(4).

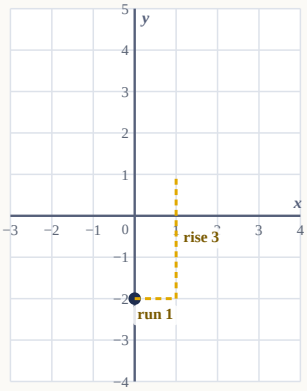


f(4) =

answer: 3

6. *Write the equation*

Write the equation of the line with slope 3 and y-intercept −2.



y =

answer: y = 3x − 2

SCORE *4.0* **Target Content**

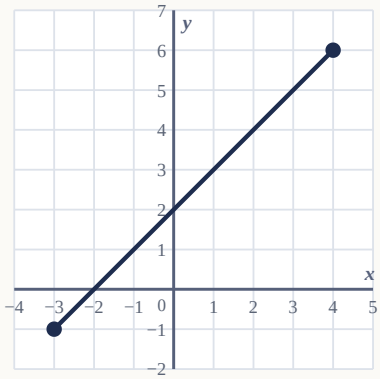
NC.M1.F-IF.5 · NC.M1.F-IF.6 · NC.M1.F-IF.7 · NC.M1.F-LE.5 · NC.M1.A-CED.2 Find domain, range, intercepts, and rates of change; write linear equations.

• *I can find domain and range, intercepts, and average rate of change, and write the equation of a line.*

Solve each problem. Show your work on paper.

7. *Domain*

The graph has a closed dot at each end. What is the **domain**?

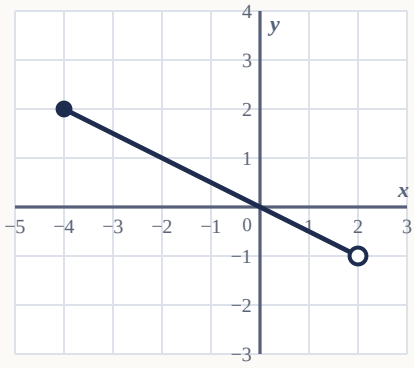


Answer:

answer: $-3 \le x \le 4$

8. *Range*

Closed dot on the left, open dot on the right. What is the **range**?

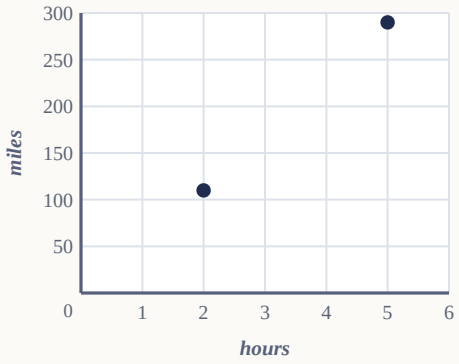


Answer:

answer: $-1 < y \le 2$

9. *Average rate of change*

A car has gone 110 miles after 2 hours and 290 miles after 5 hours. Find the average rate of change in miles per hour.



rate = _____

answer: 60 miles per hour

10. *Intercepts*

Find the **x-intercept** of 2x + 3y = 12.

x	0	?
y	?	0

Answer: _____

answer: (6, 0)

11. *Interpret*

A taxi ride costs C = 2.5m + 40 dollars for m miles. What does the ride cost for **each mile**?

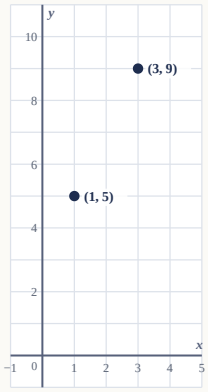
m	0	10	20
C	40	65	90

dollars = _____

answer: $2.50 per mile

12. *Write the equation*

Write the equation of the line through (1, 5) and (3, 9).



y = _____

answer: y = 2x + 3

SCORE **5.0** **Advanced Content**

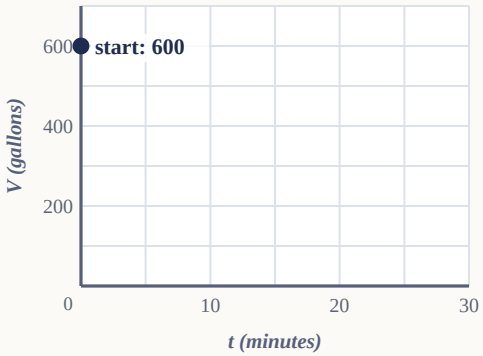
NC.M1.F-IF.2 · NC.M1.F-IF.4 · NC.M1.F-IF.5 · NC.M1.F-IF.6 · NC.M1.A-CED.2 · NC.M1.F-LE.5 Model with linear functions and explain what the numbers mean.

• *I can write a function for a situation, give a sensible domain, and explain how changing a number changes the graph.*

Solve each problem. Some answers are words.

13. *Write a function*

A pool holds 600 gallons. It drains 25 gallons each minute. Write V(t), the gallons left after t minutes.

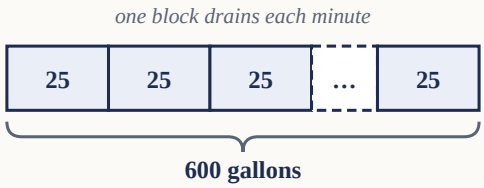


V(t) = _____

answer: V(t) = 600 − 25t

14. *Domain in context*

For the pool, V(t) = 600 − 25t. What is a reasonable **domain**? (Time starts at 0 and stops when the pool is empty.)



Answer: _____

answer: 0 ≤ t ≤ 24

15. *Use function notation*

$f(x) = 2x - 7$. Find x when $f(x) = 9$.

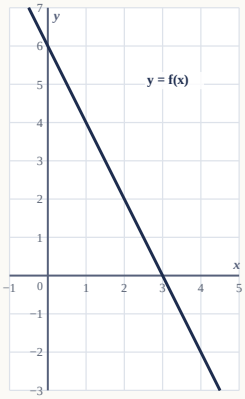


x =

answer: x = 8

16. *Positive or negative*

Use the graph of f. For which x-values is $f(x) > 0$?

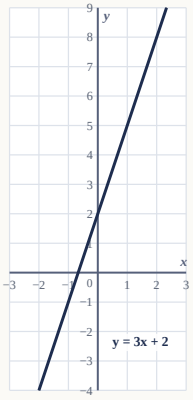


Answer:

answer: $x < 3$

17. *Change a parameter*

$f(x) = 3x + 2$ is graphed. The 2 is changed to 7. Does the graph move **up or down**, and by how many units?



Answer:

answer: up 5 units

18. *Average rate of change*

Find the average rate of change from x = 2 to x = 6.

x	0	2	4	6
y	3	4	8	15

Answer:

answer: $2.75 \left(\frac{11}{4}\right)$